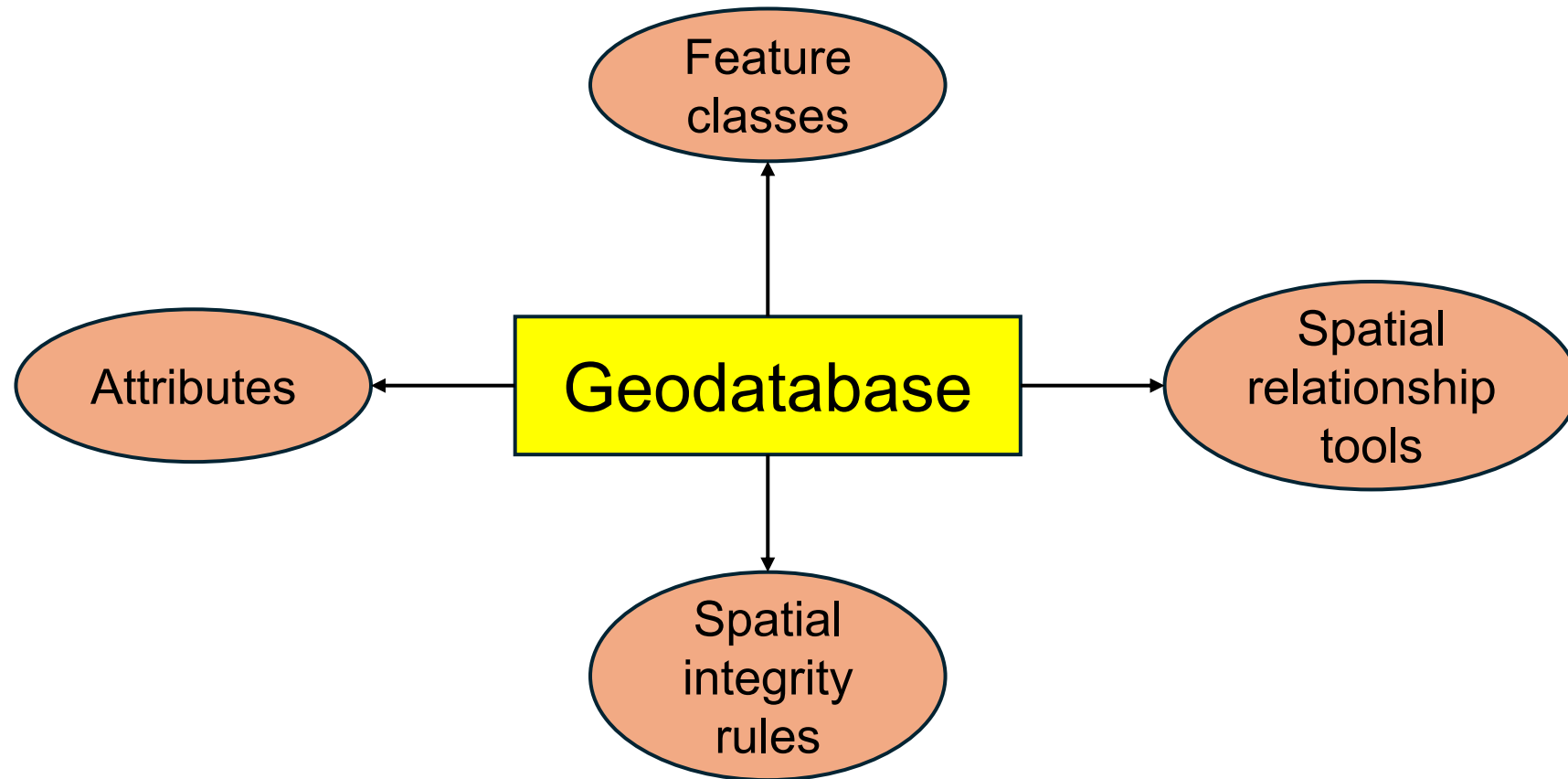


GIS Database construction,
database structure, hierarchical data,
network systems, relational database

Spatial analysis: extract, pairwise
comparison and overlay

What is a geodatabase?



What is a geodatabase? (contd.)

- Native data structure for any GIS software
- Primary format used for editing and data management
- Geographic information is stored using a database management system or file system
- Accessing and working with the dataset through a database management system using structured query language (SQL)
- GIS software logic supports working with all geographic and vector data

Common geographic data and vector file formats

- Shapefile (.shp, .shx, .dbf, .prj, .cpg)
- Computer-aided drafting – CAD (.dwg, .dxf)
- Triangulated irregular networks (.tin)
- Geography markup language (.gml)
- GeoJSON (.geojson, .json)
- GeoPackage (.gpkg)
- Keyhole Markup Language (.kml, .kmz)
- File Geodatabase (.gdb)

Database structures: Data organization in a computer

Types of data files:

- Simple lists
- Ordered sequential files
- Indexed files

Simple lists

Index		File item
Item key	Record Number	
A	1	A ₁
B	$n_a + 1$	A ₂
C	$n_a + n_b + 1$	...
...	...	B ₁
...	...	...
...	...	C ₁

Ordered sequential files

Soil profile number	Attributes					
	S	pH	De	Dr	T	E
1	A	4	deep	good	sandy	no
2	B	5	shallow	good	clay	yes
3	C	6	shallow	poor	sandy	no
4	D	7	deep	good	clay	yes
5	E	4	deep	poor	clay	no
6	F	5	shallow	poor	clay	no

S = soil series, De = depth, Dr = drainage, T = texture, E = erosion.

Indexed files

Topic	Profiles (sequential numbers in original file)					
Deep	1			4	5	6
Shallow		2	3			
Good drainage	1	2		4		
Poor drainage			3		5	6
Sandy	1		3			
Clay		2		4	5	6
Eroded		2		4		

Database structures: Data organization in a computer (contd.)

Database structures:

- Hierarchical database structure
- Network database structure
- Relational database structure
- Object-oriented database structure

Spatial analysis

- Spatial analysis allows you to solve complex location-oriented problems, explore and understand your data from a geographic perspective, determine relationships, detect and quantify patterns, assess trends, and make predictions and decisions
- Goes beyond mapping and allows you to study the characteristics of places and the relationships among them
- Combines information from many sources and derive new information by applying a set of spatial operators

Spatial analysis function classes in ArcGIS

- Geoprocessing
- Raster functions
- Charts
- Data engineering
- Artificial intelligence
- 3D analyst
- Business analyst
- Geostatistical analyst
- Image analyst
- Link analysis
- Network analyst
- Spatial analyst
- Visibility analyst

Geoprocessing

- Framework and set of tools for processing geographic and related data
- Performs spatial analysis and manages GIS data in an automated way

Analysis toolbox

- Performs analytical operations such as spatial overlays, buffer creation, calculating statistics, proximity analysis

Extract

Clip: Extracts input features that overlay clip features

Select: Extracts features from an input feature class and stores them in an output feature class

Split: Splits an input with overlaying features to create a subset of output feature classes

Analysis toolbox (contd.)

Pairwise comparison

- Provides a framework to derive relative weights of importance for multiple variables
- Provides a structured method to assess the importance of the variables by comparing them two at a time, instead of simultaneously ranking all the variables
- Useful when dealing with intangible factors that cannot be measured directly but are essential to decision making
- Relies on expert judgments to derive priority scales that measure the factors in relative terms

Analysis toolbox (contd.)

Example applications of pairwise comparison

- Calculate the weights for multiple criteria in a suitability model.
- Determine the weights for a set of submodels for a composite suitability model.
- Assess the relative importance of different land use types within a layer.
- Compute the relative weights of different pollutants to create an air quality index to influence public policy.
- Identify new class values when reclassifying a raster layer based on preference.
- Estimate the relative priorities for multiple projects in business decision making.

Pairwise comparison (contd.)

Geoprocessing Assign Weights By Pairwise Comparison

Parameters Environments

Input Variables To Compare

- Solar_Gain
- Aspect
- Elevation
- Dist_to_Road
- Dist_to_Elect

Output Table: OutWeightTable

☒ Add comparison matrix to output

Pairwise Comparison Matrix

Define Pairwise Comparisons

Importance: Extreme (9, 7), Strong (5, 3), Equal (1, 3), Strong (5, 7), Extreme (9)

Comparison Matrix:

Variables	Solar_Gain	Aspect	Elevation	Dist_to_R	Dist_to_E	Weights
Solar_Gain	1	3	2	2	0.333	0.219
Aspect	0.333	1	0.5	1	0.2	0.082
Elevation	0.5	2	1	1	0.5	0.148
Dist_to_R	0.5	1	1	1	0.2	0.104
Dist_to_E	3	5	2	5	1	0.447

Consistent Ratio: 0.029101

Buttons: Consistency, OK, Cancel

Run

Comparison matrix

Variables	Solar_Gain	Aspect	Elevation	Dist_to_Road	Dist_to_Elect
Solar_Gain	1	3	2	2	0.333
Aspect	0.333	1	0.5	1	0.2
Elevation	0.5	2	1	1	0.5
Dist_to_Road	0.5	1	1	1	0.2
Dist_to_Elect	3	5	2	5	1

Pairwise comparison (contd.)

Pairwise comparison uses a 1 to 9 scale developed by Saaty (2008) with their scales, definitions, and explanations of each value in the scale

Intensity of importance	Definition	Explanation
1	Equal importance	Two activities contribute equally to the objective.
2	Weak or slight importance	
3	Moderate importance	Experience and judgement slightly favor one activity over the another.
4	Moderate plus importance	
5	Strong importance	Experience and judgement strongly favor one activity over another.
6	Strong plus importance	
7	Very strong or demonstrated importance	An activity is favored very strongly over another; its dominance is demonstrated in practice.
8	Very, very strong importance	
9	Extreme importance	The evidence favoring one activity over another is of the highest possible order of affirmation.

Analysis toolbox (contd.)

Overlay toolset

Consists of tools to overlay multiple feature classes to combine, erase, modify, or update spatial features, resulting in a new feature class

- Apportion polygon
- Counting overlapping features
- Erase, identify, intersect, update
- Spatial join, remove overlap, union
- Symmetrical difference

Analysis toolbox (contd.)

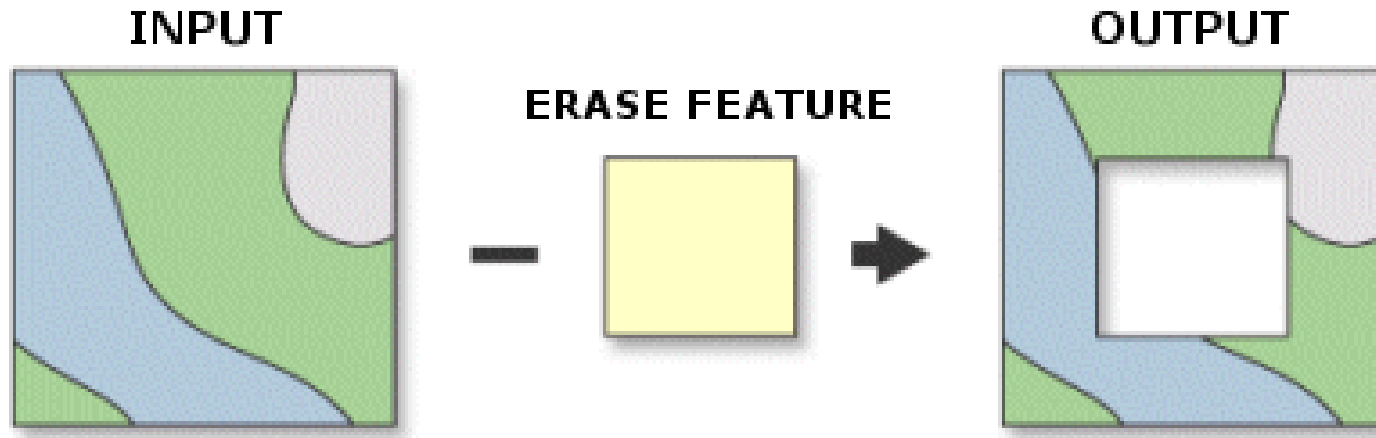
Overlay toolset

- **Apportion polygon:** Summarizes the attributes of an input polygon layer based on the spatial overlay of a target polygon layer and assigns the summarized attributes to the target polygons. The target polygons have summed numeric attributes derived from the input polygons that each target overlaps. This process is typically known as apportioning or apportionment.
- **Count overlapping features:** Generates planarized overlapping features from the input features. The count of overlapping features is written to the output features.

Analysis toolbox (contd.)

Overlay toolset

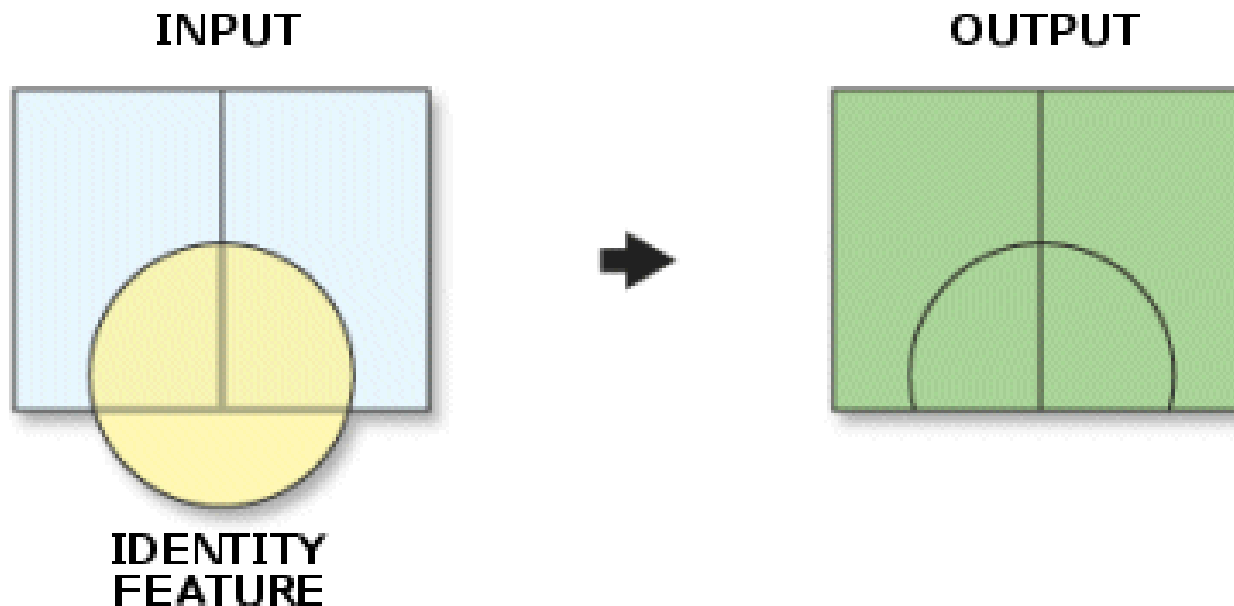
- Erase: Creates a feature class by overlaying the input features with the erase features. Only those portions of the input features falling outside the erase features are copied to the output feature class



Analysis toolbox (contd.)

Overlay toolset

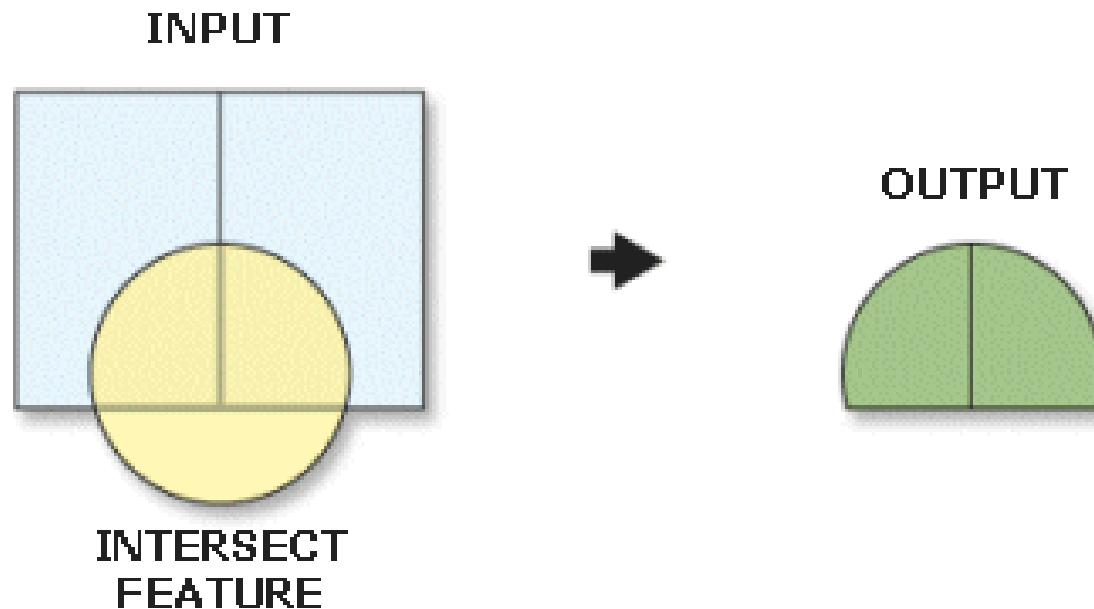
- Identity: Computes a geometric intersection of the input features and identity features. The input features or portions thereof that overlap identity features will get the attributes of those identity features.



Analysis toolbox (contd.)

Overlay toolset

- Intersect: Computes a geometric intersection of the input features. Features or portions of features that overlap in all layers or feature classes will be written to the output feature class.



Analysis toolbox (contd.)

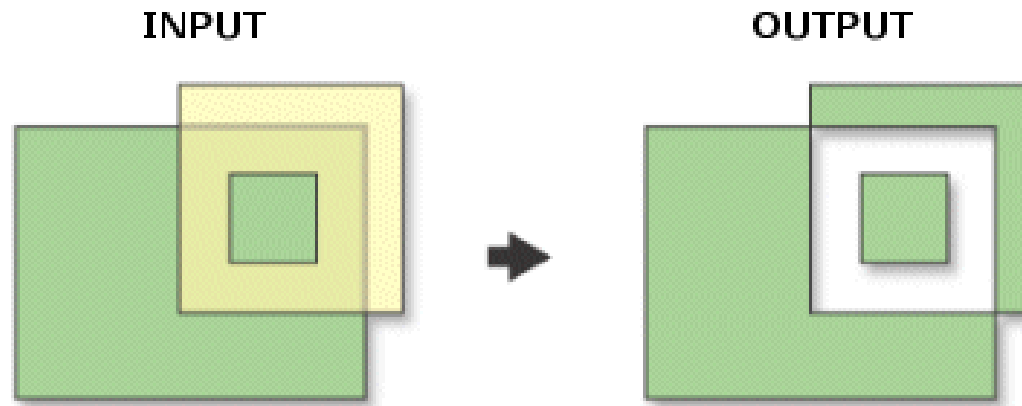
Overlay toolset

- Remove overlap: Removes overlap between polygons contained in multiple input layers.
- Spatial join: Joins attributes from one or more inputs to another input based on the spatial relationship. The target features and the joined attributes from the join features are written to the output feature class.

Analysis toolbox (contd.)

Overlay toolset

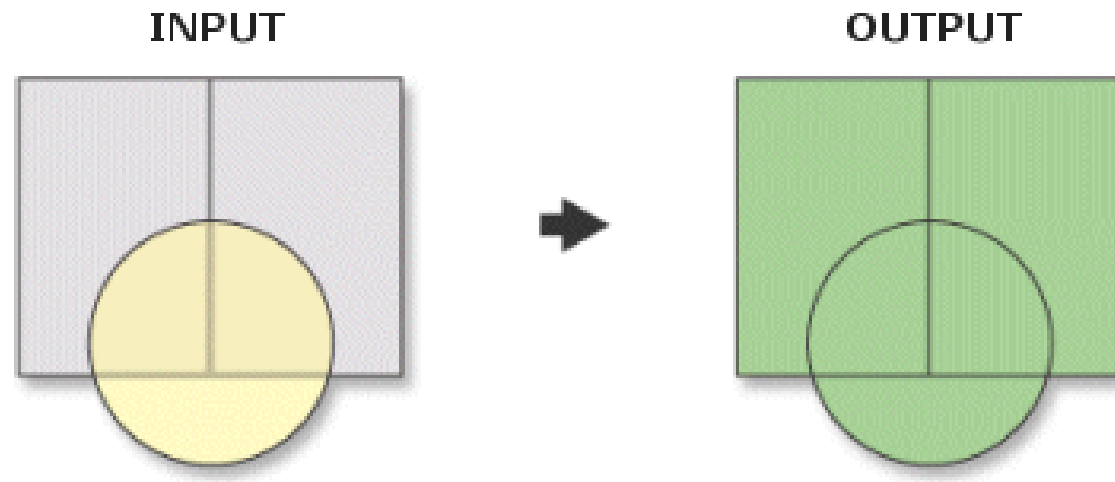
- Symmetrical difference: Computes a geometric intersection of the input and update features, returning the input features and update features that do not overlap. Features or portions of features in the input and update features that do not overlap will be written to the output feature class.



Analysis toolbox (contd.)

Overlay toolset

- Union: Computes a geometric union of the input features. All features and their attributes will be written to the output feature class.



Analysis toolbox (contd.)

Overlay toolset

- Update: Computes the geometric intersection of the input features and update features. The attributes and geometry of the input features are updated by the update features in the output feature class.

